

Energy Isolation Safety Standard

CLEVELAND-CLIFFS STEEL

Energy Isolation Safety Standard

Effective Date:	<u>9-1-2022</u>	Supersedes:	<u>All Previous Versions</u>
		Standard No.:	<u>CCS HSS STD 001.0</u>
		Follow-Up Review Date:	<u>9-1-2023</u>

APPROVED

K. Dirk Berard Senior Director Health Safety Security	<u>8/31/2022</u>
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APPROVER

DATE

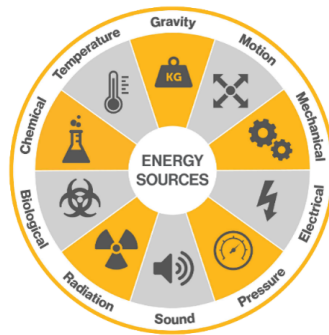
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Energy Isolation Safety Standard

1 Purpose/Scope

- 1.1 This Energy Isolation Safety Standard outlines the responsibilities under this standard, minimum verification procedure, training, and periodic inspection requirements to protect people from potentially hazardous energy when servicing or performing maintenance on equipment, industrial vehicles, and machinery.
- 1.2 When discussing hazardous energy, we mean any energy that can be released and might harm a person, such as:

Chemical
Temperature
Gravity
Motion
Mechanical
Electrical
Pressure
Sound
Radiation
Biological



- 1.3 These are examples of hazardous energy being released and causing harm.
- While workers are repairing a connection in piping, somewhere upstream of that repair, another worker opens a valve sending fluids down the pipe that spills on and burns the maintenance workers.
 - When a conveyor jams, a worker reaches in to try to clear the jam, when the jam is suddenly and unexpectedly freed, the worker is crushed as a result.
 - While a worker is servicing a machine, the internal wiring in the machine shorts, which causes the maintenance worker to be shocked.
- 1.4 Energy Isolation ensures these types of hazardous energy sources are isolated, controlled, or rendered in a zero-energy state to perform servicing or maintenance safely.
- 1.5 This Energy Isolation Safety Standard applies to all Cleveland-Cliffs Steel employees, contractors, sub-contractors, service providers, vendors, and visitors when:
- Performing servicing and maintenance on equipment, industrial vehicles, or machines in which the energization or startup of the machines or equipment, or the release of stored energy, could harm employees regardless of the length of the exposure.

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- Servicing and maintenance require the removal of machine guards or bypassing other safety devices.
- Servicing and maintenance require personnel to place any part of their body into a point of operation, an area on a machine, a piece of equipment where work is performed, or into the danger zone associated with the machine's operation.

1.6 This Energy Isolation Safety Standard does NOT apply to:

- Work on cord and plug connected electrical equipment if the equipment is unplugged from the energy source, and the authorized employee has exclusive control of the plug.
- Hot tapping operations.
- Energized electrical work for purposes of diagnostic work.
- Minor tool changes and adjustments and other minor servicing activities during normal production operations.

2 Responsibilities and Authorities

2.1 This standard was jointly developed with participation from the represented workforce and management.



2.2 Cleveland-Cliffs Steel is responsible for:

- Providing this This Energy Isolation Safety Standard.
- Complying with the various provisions of Federal, State, and Local rules and regulations.

2.3 Plant General and Health and Safety Managers are responsible for:

- Establishing and maintaining guidance and implementation documents in compliance with this Standard.
- Implementing the safety inspection process for new or modified equipment.
- Audit for compliance with this Standard.
- Ensuring lockout/tryout initial, annual and refresher training are available and conducted in compliance with this Standard.
- Investigating all lockout/tryout near misses, incidents, and injuries.

2.4 Division Managers are responsible for:

- Ensuring that Section and Shift Managers are trained on all aspects of this Energy Isolation Standard.
- Ensuring employees attend energy Isolation training.
- Investigating all lockout/tryout near misses, incidents, and injuries.
- Ensuring that each energy isolation verification procedure receives a periodic inspection annually.

2.5 Section Managers (Operations and Maintenance) are responsible for:

- Ensuring correct lockout devices are available to control hazardous energy successfully.
- Ensuring that the energy isolation verification procedures are developed and maintained for jobs/tasks in their area of responsibility.
- Authorizing new or changed energy isolation verification procedures and compliance with this Standard.
- Ensuring new or transferred employees receive hands-on training.
- Ensuring employees attend energy isolation training.
- Investigating all lockout/tryout near misses, incidents, and injuries.

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2.6 Shift Managers (Operations and Maintenance) are responsible for:

- Ensuring employees are trained on applicable energy isolation and verification procedures.
- Shift Managers authorize group lockout leaders utilizing the energy isolation Authorized Employee Evaluation Form (CCS-HSS-FRM-0003.0).
- Ensuring group and individual lockouts are performed when required.
- Investigating all lockout/tryout near misses, incidents, and injuries.
- Ensuring that employees, contract personnel, and visitors comply with the requirements detailed in this Standard.

2.7 Employees are responsible for:

- Reviewing and following energy isolation and verification procedures.
- Signing on to a group lockout box control list.
- Applying your personal safety lock to energy isolation devices, hasp, or a group lockout box.
- Personal safety lock removal when appropriate.
- Informing your manager if you have a concern with an energy isolation activity.
- Compliance with the use of three-part communications.
- Compliance with this Standard.

2.8 A Group Lockout Leader is:

- Experienced and knowledgeable of the machine/equipment operation.
- Capable of recognizing hazardous energy sources, the type and magnitude of the energy available, and the methods and means necessary for energy isolation.
- An authorized employee selected by their Shift Manager who meets the Group Lockout Leader evaluation criteria outlined in Form CCS-HSS-FRM-0003.0.
- The above evaluation is one and done unless a periodic inspection, observation, or interaction reveals deviations or inadequacies in the employee's knowledge of Group Lockout Leader duties.
- Trained to initiate energy isolation activities.
- Trained to apply locks to energy isolation devices, perform second-person verification and try out equipment,
- Trained to apply the control lock for the group lockout box and monitor completion of the group lockout box control list,
- Not a permanent position unless the department makes it a permanent position.
- Responsible for assessing the equipment, facilitating the group energy isolation verification procedure, maintaining continuity throughout the energy isolation and activity, and ensuring safe re-energization of the machine or equipment.

2.8.1 It is essential to understand that the same person cannot apply the locks to energy isolation devices and verify the locks - Second Person Verification is required!

2.8.2 Managers are to follow the training and retraining outlined in this document.

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2.9 A Second Person Verifier is:

- Experienced and knowledgeable of the machine/equipment operation.
- Capable of recognizing hazardous energy sources, the type and magnitude of the energy available, and the methods and means necessary for energy isolation.
- An employee selected and trained by their Shift Manager, to perform the duties outlined in this document.
- Managers are to follow the training and retraining outlined in this document.
- Responsible for walking down each assigned energy isolation device required to be locked out according to the energy isolation verification procedure to assure equipment lock(s) were applied and secured to all identified energy isolation devices (Verify the Locks).
- Not a permanent position unless the department makes it a permanent position.
- Second Person Verification is required when multiple people, multiple energy sources, or group lockout boxes are used.
- Lockout involving just one authorized individual for a single energy source requires a "Tryout" of the equipment but does not require second-person verification.

2.9.1 It is essential to understand that the same person cannot apply the locks and verify the locks - Second Person Verification is required!

2.10 A Tester/Tryout Person is:

- Experienced and knowledgeable of the machine/equipment operation.
- Capable of recognizing hazardous energy sources, the type and magnitude of the energy available, and the methods and means necessary for energy isolation.
- An employee selected and trained by their Shift Manager to perform the duties outlined in this document.
- Responsible for ensuring the area/equipment cleared of workers, the area cleared of tools/materials, machine guards reinstalled and secured, lockout device(s) removed, informs affected employees, operates equipment (Tries Out), and returns controls to the "off" or "neutral" position after the tryout.
- Not a permanent position unless the department makes it permanent.

2.11 Group Lockout Leaders, Second Person Verifiers, and Testers can be cross-trained.

2.12 There shall be a master list maintained in each division of these authorized employees trained for group lockout leaders, second-person verifiers, and testers. Use form (CCS-HSS-FRM-0004.0, energy isolation Training Log).

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- 2.12.1 Three-person example: Employee A (Group Lockout Leader) applies the locks, and Employee B (Second Person Verifier) walks down the unit and verifies the locks. Then, Employee C (Tester) tries out the equipment.
- 2.12.2 Two-person example: Employee A (Group Lockout Leader) applies the locks, and Employee B (Second Person Verifier) walks down the unit and verifies the locks. Then, Employee A (Group Lockout Leader) or Employee B (Second Person Verifier) tries out the equipment.

2.13 Responsibility for Lock Placement and Lock Removal

- 2.13.1 The employee placing a personal safety lock on the energy isolation device is responsible for removing that lock. If the service being performed on the equipment has not been completed at shift change, or if the employee is called off the job, that employee must remove their personal safety lock(s). The incoming individual continuing the work must ensure that all energy sources are secured and properly locked out.
- 2.13.2 If it is necessary to cut off a personal safety lock that an employee or contract employee failed or forgot to remove, approval must come from the Division Manager. A Safety Lock Removal Incident Report Form must be completed.
- 2.13.3 A reasonable effort shall be made to contact the employee and ensure they are not at the worksite. This effort shall consist of:
- Making phone or radio contact.
 - Checking clock-out records and sign-in sheets.
 - Thoroughly inspect the entire area to ensure the employee is not present.
 - If necessary to cut the personal safety lock, the employee must be notified before returning to work.
- 2.13.4 Each employee is responsible for placing their personal safety lock on the correct energy isolation device or group lockout box.
- 2.13.5 The ultimate responsibility to recognize, lock, try and ensure all hazardous energy sources lie within each person who might be required to apply or abide by the provisions of this Standard.

3 References and Definitions

3.1 References

- 29 CFR 1910.147 OSHA (Occupational Safety & Health Administration) Lockout Standard
- ANSI (American National Standards Institute) Z244.1-2016, the Control of Hazardous Energy Lockout and Alternative Controls

3.2 Definitions

Alternative Controls (under the minor tool changes, adjustments, and minor servicing exemption): The means of controlling hazardous energy (other than energy isolation) to reduce risk to an acceptable level.

- Alternative controls may include devices such as but are not limited to engineered systems such as light curtains, pressure mats, area scanners, hold-to-run devices, interlock switches, stop devices, trapped key devices, specially designed tools, remote devices, interlocked barrier guards, local disconnects or keyed control switches, restraint devices that prevent the employee from entering the zone of danger, and Plexiglas or other physical and substantial protective barriers in a fixed position preventing the employee's body from entering the machine hazard area and adjacent equipment that could cause harm.
- Before energy isolation is implemented on a unit, all **alternative control** devices shall be checked and verified that the control keys, switches, or interlocks are in their proper position and the unit is in normal operations mode. This ensures that no alternative controls interrupt normal operation and do not interfere with the energy isolation and verification.

Authorized Employee: An employee who locks out machines or equipment to perform servicing or maintenance.

Affected Employee: An employee who is required to use machines or equipment on which servicing or maintenance is being performed under the energy isolation standard or who performs other job responsibilities in an area where such servicing or maintenance is performed. An **affected employee** becomes an **authorized employee** when that employee's duties include servicing or maintenance.

Blind: Is the absolute closure of a pipe, line, or duct achieved by fastening a solid plate, threaded plug, or cap across its bore to cover it completely. The cover must:

- A. At least cover the outer edge of a flange's mating surface.
- B. Be capable of withstanding the maximum upstream system pressure during normal operations.
- C. Blinds include *blanks, slip plates, blind flanges, threaded caps, and physical disconnects*.
- The three types of blinds utilized include:
 - A. Isolation
 - B. Hydrotest
 - C. Permanent Blinds must be capable of holding the maximum pressure and temperature to which the blind will be exposed.

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- Examples of acceptable handles are the "Spectacle," "T handle," or "J handle" design.
- To eliminate the possibility of confusing restriction orifices with blinds, blinds with straight tabs must not be used.



Figure 1 (above) - Blind-Flange Lockout Device

- The crew securing the pipeline locks out upstream valves and then places a blank or blind in the system to positively block any potential downstream flow.
- The flange bolts are then torqued to proper specification. At least one torqued bolt is covered with the blind-flange lockout device (see Figure 1 above).
- The device is secured with one or more personal safety locks, or a group lockbox could be used. Multiple bolts could be covered using multiple devices if desired.

Capable of Being Locked Out: An **energy-isolating device** is considered capable of being locked out if:

- It is designed with a **hasp** or other means of attachment to which a lock can be affixed.
- It has a locking mechanism built into it.
- It can be locked without dismantling, rebuilding, replacing the **energy-isolating device**, or permanently altering its energy control capability.

Control Lock (Group Lockout Box): A yellow, keyed lock having only one key is used to secure the **group lockout box** to provide continuity of a lockout for multiple energy sources, multiple crews, or across shifts.

- **Control locks** shall be engraved with the division's name and "Control Lock."

Danger Zone: The danger zone is the machine's point of operation area from which the operator must stay clear during the machine's operation cycle to prevent injury (caught in, burn, shock, etc.).

Energized: Machines and equipment are energized when connected to an energy source or when they may contain residual or stored energy.

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De-Energized: Disconnected from all energy sources and not containing residual or stored energy.

Double Block and Bleed: Closing a line, duct, or pipe by closing and locking two inline valves and opening and locking a drain or vent valve in the line between the two closed valves.

Energy-Isolating Device: A mechanical device that physically prevents the transmission or release of energy, including but not limited to the following:

- A manually operated electrical circuit breaker.
- A disconnect switch.
- A manually operated switch by which the conductors of a circuit can be disconnected from all ungrounded supply conductors, and in addition, no pole can be operated independently.
- A valve.
- A block and any similar restraint device are used to block or secure stored energy sources or prevent equipment and machinery from accidental movement.
- **NOTE:** Push buttons, selector switches, wall switches, and other control circuit type devices are **not energy isolating devices**.
 - A. All **energy isolation devices** shall be labeled or marked to indicate their function. The identification shall include the following:
 1. Machine, equipment, or process supplied; and
 2. Energy type and magnitude.
 - B. The marking shall be of sufficient durability to withstand the anticipated environment.
 - C. The potential for injury and incidents may be reduced if employees are not expected to rely on memory as to which isolating devices apply, particularly in complex equipment. There is a greater potential for error when unlabeled or inadequately identified energy-isolating devices exist.
 - D. Examples of marking and labeling include attached or embossed markings and signs such as:
 - “Main Breaker (480 V) Press 3”
 - “Natural Gas Shutoff (20 psi) Process Line 2”
 - “Drive Power Isolation (120V), Axis X, Only”
 - “High Pressure (600 psi) Return Line”
 - E. Electrical boxes must be labeled directly on the box. Valves may be labeled on the valve body or a suspended durable sign or tag. Stencil, color, tags, or other methods may be labeled by constraints, blocks, chains, or pins.

Energy Source: Any chemical, temperature, gravity, motion, mechanical, electrical, pressure, sound, radiation, biological, or other energy sources.

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Equipment Lock: A blue, keyed lock having only one key is used to secure an **energy isolation device** on machines or equipment.

Equipment locks shall be marked with terminology matching the **energy isolation device** (e.g., #5 Scrap Baler Hydraulic Supply Valve).

- Because **energy isolation device** terminology can be lengthy, it is permitted to number **equipment locks** sequentially if a legend exists describing each **equipment lock**, and a legend is provided with each energy isolation.
- The energy isolation verification procedure can be the legend.
- In place of marking equipment locks, a durable tag with a grommet can be attached to the shank of an equipment lock to identify the equipment lock.

Grounding: Conducting connection between an electrical circuit or equipment and the earth.

Group Lockout: A lockout that occurs when one or more employees perform servicing or maintenance on equipment with multiple energy sources simultaneously.

Group Lockout Box: A storage device with a see-through front used in group lockout to secure all keys to the equipment locks secured on an energy isolating device(s) and permits all exposed persons to apply their personal safety lock.

Hazardous Energy: Any chemical, temperature, gravity, motion, mechanical, electrical, pressure, sound, radiation, and biological or other energy can harm personnel. Some energy sources are apparent such as electricity, heat in a furnace, or something that might fall. Others may be hidden hazards such as air pressure in a system or a tightly wound spring.

Hot tapping or pressure tapping connects existing piping or pressure vessels without interrupting or emptying that section of pipe or vessel. As a result, the pipe or tank can continue operating while maintenance or modifications are being done. The process is also used to drain off pressurized casing fluids and add test points or sensors such as temperature and pressure.

- Hot taps can range from a ½-inch hole designed for something as simple as quality control testing to a 48-inch tap to install a variety of ports, valves, t-sections, or other pipes.
- Hot tap operations that involve transmission and distribution systems for gas, steam, water, or petroleum products on pressurized pipelines, if continuity of service is essential, the shutdown of the system is impractical, documented procedures are followed, and employees are effectively protected by special equipment.

Idled Equipment: Equipment no longer in use or will be out of service beyond a month. This equipment requires that a green lock be applied.

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- Do Not Operate (DNO) tags can no longer be used as the sole means for designating out-of-service equipment.

Industrial Vehicles: includes all powered industrial trucks (PIT), fork trucks, tractors, platform lift trucks, trucks, cars, motorized personnel carts, material handling wheeled or tracked industrial vehicles, and other specialized industrial trucks powered by electric motors or internal combustion engines used exclusively in the industrial setting. The term "Industrial Vehicle" will mean all the above.

Lockout: The placement of a **lockout device** on an **energy-isolating device** so that the **device** and the equipment being controlled cannot be operated until the **lockout device** is removed. Lockout is not completed and ready for individuals to add their safety lock to the lock box until verification and try-out are completed and recorded on the energy isolation and verification procedure.

Lockout Device: Any device that uses a positive means such as blank flanges, bolted slip blinds, and other devices to hold an **energy-isolating device** safely before a standard lock is applied, thereby preventing the energizing of machinery or equipment.

- See Figures 2, 3, and 4 below for other lockout devices (e.g., electrical, motor vehicle, valve, cable):

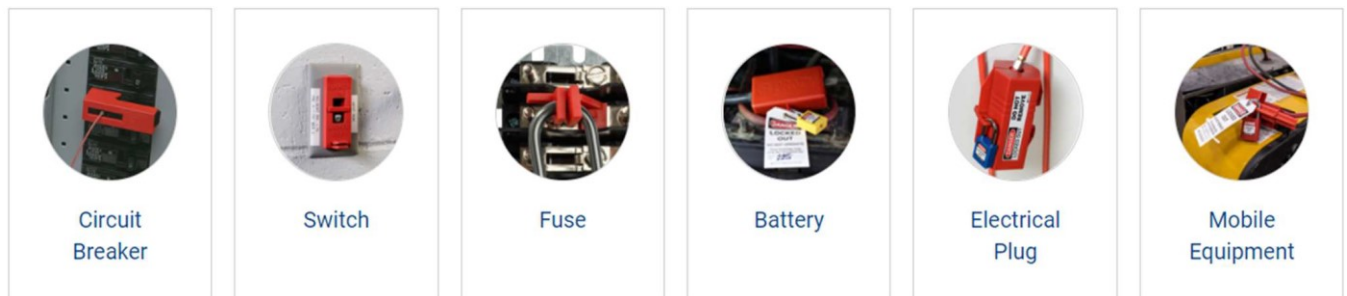


Figure 2 (above) - Example of Electrical and Industrial Vehicle Lockout Devices

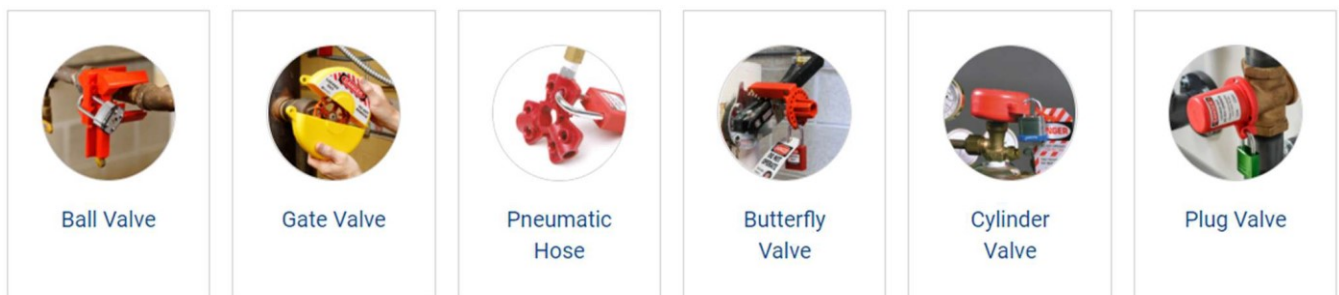


Figure 3 (above) - Example of Valve Lockout Devices

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Figure 4 (above) - Example of Cable Lockout Devices

Lockout Hasp: A device that provides effective multi-person lockout on an **energy isolation device** or **group lockout box** (see Figure 5 below). Multiple **personal safety locks** can be applied to a lockout hasp, allowing a single **energy isolation device** to be isolated by more than one worker or multiple **personal safety locks** to a **group lockout box**.



Figure 5 – Example of Lockout Hasp

Energy Isolation) Verification Procedure: A written, standardized and controlled procedure prepared for a piece of equipment or process system detailing the correct lockout/tryout verification process for **hazardous energy**, including residual and stored energy (see form CCS-HSS-FRM-0001.0).

Normal Production Operations: Utilize a machine or equipment to perform its intended production function.

Other Employees: All employees who may be in an area **where energy isolation and verification procedures** may be utilized.

Personal Safety Lock: A red, keyed lock having only one key and used only by the trained and authorized employee; the lock was assigned to ensure personal protection during lockout/tryout energy isolation activities.

- Personal safety locks must be marked with the employee's name, payroll, clock, or badge number.

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Plug-Controlled Equipment: Cord and plug connected electric equipment for which exposure to the hazards of energization or startup of the equipment is controlled by unplugging the equipment from the energy source.

Point-of-Operation: The area on a machine where work is performed upon the material being processed. The point of operation of machines whose operation exposes an employee to injury shall be guarded.

Second-Person Verification: A second authorized individual (the Verifier) shall verify that each equipment lock is placed and secured correctly on each energy isolation device.

- Second Person Verification is required when multiple people, multiple energy sources, or group lockout boxes are used.

A. "Lockout-***Verify the Locks***-Tryout."

- Lockout involving just one authorized individual for a single energy source requires a "Tryout" of the equipment but does not require second person verification or an energy isolation and verification procedure.

Servicing and Maintenance: Workplace activities such as constructing, installing, setting up, adjusting, inspecting, modifying, maintaining, and servicing machines or equipment, including lubrication, cleaning or unjamming of machines or equipment, and adjusting or tool changes where employees could be exposed to the energization or startup of the equipment or release of ***hazardous energy***.

Setting Up: Any work performed to prepare a machine or equipment to perform its ***normal production*** operation.

Under Exclusive Control: The ***authorized employee*** would be able to prevent the equipment from becoming reenergized during their servicing or maintenance of the equipment. ***Plug-controlled equipment*** is under the ***exclusive control*** of the employee if the plug is physically in possession of the employee, within arm's reach and in line of sight of the employee, or if the employee has affixed a ***personal safety lock*** on the plug.

Tags for Tagout: The placement of a tag only on an ***energy-isolating device*** is prohibited for energy isolation unless the energy isolation device does not accommodate a lock (see Section 21).

Tags for Lock Identification:

- Substantial tags with grommets can be used on contractor personal safety locks for outages and travelers.
- Tags can be attached to the shank of equipment locks if you choose not to engrave the equipment locks. The markings on the tag or equipment lock must be legible and match the identification of the energy isolation device.

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Three-Part Communication: Anytime more than one person or crew is involved in a task that requires understanding and coordination between the two, clear and concise Initiate-Acknowledge-Confirm (IAC) communication shall be utilized.

- A key to excellent communication is establishing procedures and a culture that utilizes Initiate-Acknowledge-Confirm 100% of the time.
- With three-part communication, a mishap can be avoided:

Employee A – entry end operator:

“Are your hands out of the way?”

Employee B – removing equipment locks after a maintenance turn:

“My hands are clear; go ahead and bump the mandrel rewind motor.”

Employee A – entry end operator:

“Your hands are clear; now bumping the mandrel rewind motor.”

Tryout: Tryout requires that you physically attempt to turn on all power switches and devices once the equipment has been locked out. The tryout is your final check and assurance that the equipment has been isolated from all power sources. Tryout can be accomplished in the following ways:

- Attempting to operate the equipment (manually shifting operating valves, turning on switches, etc.).
- Visually verifying that needed safeguards are in place or that blades in the corresponding disconnect box are disengaged.
- Using a voltmeter or other test device.
- Some lockout/tryout activities will require all three tryout methods to ensure that **energy sources** are properly locked out.

Zero Energy State: means a condition reached when all energy sources to or within equipment are isolated, blocked, or otherwise relieved with no possibility of re-accumulation.

4 Resources

4.1

5 Requirements

5.1 Cleveland-Cliffs shall provide training to ensure that employees understand the purpose and function of the lockout/tryout energy isolation standard and that the knowledge and skill required for the safe application, usage, and removal of the energy isolations are acquired by employees.

5.2 The Health, Safety, and Security Manager or designee will conduct initial training during new hire orientation.

5.3 Next, hands-on training will be conducted by the employee's Shift Manager and trained co-workers.

5.3.1 This Energy Isolation Safety Standard will be presented at least annually during a safety meeting by an authorized manager.

- Refresher training shall also occur when a periodic inspection reveals deviations or inadequacies in the employee's knowledge or use of the energy isolation verification procedure(s).

5.4 Training shall include the following:

5.4.1 Authorized Employees – *Training* in recognizing applicable hazardous energy sources, the type and magnitude of the energy available in their workplace, and the methods and means necessary for energy isolation.

5.4.2 Affected Employees – *Training* will include the purpose and use of the energy isolation verification procedures.

5.4.3 Other Employees – *Training* will include instruction on the energy isolation verification procedures and the prohibition relating to restarting or reenergizing machines or equipment which are locked out.

5.5 Employee retraining is required:

5.5.1 Whenever an employee is transferred, or there is a change in their job assignments (a position change that requires a new job orientation).

5.5.2 Whenever a new hazard is introduced due to a change in machines, equipment, or process.

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5.5.3 Whenever there is a change in energy isolation verification procedures.

5.5.4 Whenever a periodic inspection reveals inadequacies in the company's energy isolation verification procedures or the employee's knowledge of the energy isolation verification procedures.

5.6 This Energy Isolation Safety Standard shall be covered with employees yearly in a safety meeting(s).

5.6.1 Training must be documented with the employee's printed name, signature, employee ID number, date of training, and the printed name and signature of the person conducting the training. Use Training Log (CCS-HSS-FRM-0004.0).

5.7 Seven Steps to Safe Equipment Shutdown

- The following procedure shall be utilized to lockout and try out hazardous energy sources to achieve a zero-energy state when performing servicing and maintenance so that maintenance and servicing can be performed safely.

5.7.1 **Hazardous Energy Assessment** – Before beginning work on equipment, perform a TAKE 5 to assess the equipment to determine the energy sources and the appropriate energy isolating devices.

- Assess potentially hazardous stored or residual energy and initiate measures to relieve, disconnect, restrain or otherwise render “safe” any stored or residual energy. Ensure re-accumulation of stored energy is not possible.
- Consult the applicable energy isolation verification procedure.

5.7.2 **Notification** – Notify all affected employees that an energy isolation procedure will be utilized. Do not proceed with the lockout until authorized by the equipment operator(s).

5.7.3 **Shutdown** – Shut down or turn off the machine or equipment using normal or emergency shutdown procedures established for the machine or equipment.

5.7.4 **Isolate** – Physically locate and isolate all energy isolating devices needed to control the energy to the machine or the equipment.

5.7.5 **Lockout** – Authorized employees shall place and secure applicable lockout devices to each energy-isolating device identified in the energy isolation and verification procedure. Place your personal safety lock on the lockout device(s), locking them in a “safe” or zero energy position. In the case of a group lockout, place your personal safety lock on the group lockout box.

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5.7.6 *Second Person Verification* – Physically walk down each energy isolation device to verify that an equipment lock(s) is applied and secured correctly to each energy isolation device.

- Second Person Verification is required when multiple people, multiple energy sources, or group lockout boxes are used.
- Lockout involving just one authorized individual for a single energy source requires a “Tryout” of the equipment but does not require second-person verification.
- Second Person Verification can be accomplished simultaneously after the equipment locks are placed and secured.

5.7.7 *Tryout* – Before starting servicing or maintenance, ensure all employees are in a safe location. Tryout can be accomplished through the following acts:

- Attempting to operate the equipment.
- Visually verifying that needed safeguards are in place or that blades in the corresponding disconnect box are disengaged.
- Using a voltmeter or other test device.
- Some lockout/tryout activities will require all three tryout methods to ensure that **energy sources** are properly locked out.

NOTE: If the machine/equipment operates or is not in a zero-energy state, notify your Shift Manager immediately and do not service or perform any work tasks on the equipment until isolation can be verified.

5.8 Seven Steps to Safe Equipment Start-Up

- The procedure listed below shall be utilized to return lockout equipment to normal operation safely.

5.8.1 *Inspect* – Inspect the work area for personnel, ensure all guards have been reinstalled, all tools and materials have been removed, and the equipment is ready for normal operation.

5.8.2 *Notify* – Notify all affected employees and operator(s) in the work area that the equipment will be reenergized.

5.8.3 *Complete* – Complete the Post energy isolation activity checklist.

5.8.4 *Confirm* – Confirm that all employees are in a safe location and clear from the equipment.

5.8.5 *Remove* – Remove your lock and lockout hasp from the energy isolation device(s) or group lockout box or any blanks or blinds on piping.

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- 5.8.6 *Restore* – Restore energy to the machine/equipment. Visually verify that the area is clear of personnel and tools and that guards are in place and secure before restoring energy.
- 5.8.7 *Inform* – Inform all employees and operators that the equipment is ready for normal operation.

5.9 Standardized Lock, Lock Purpose, and Lock Identification

- RED = **Personal safety locks** (Cliffs and Contractor)
- BLUE = **Equipment locks**
- YELLOW = **Control and cross locks** for group lockout boxes
- GREEN = **Idled equipment locks** (green locks are not to be used for protecting personnel performing servicing or maintenance)

5.9.1 The standard lock for all Cleveland-Cliffs Steel:

- An American Standard keyed lock.
- Locks can have either 1-1/2 inch or 3-inch shanks.
- Only used to secure hazardous energy sources (except green), and those locks shall not be used for any other purpose.

5.9.2 There shall be no duplicate or additional keys for personal safety, equipment, or control locks.

5.9.3 Any secondary keys must be destroyed.

5.9.4 All attempts will be made to eliminate the ability to order locks with more than one key through procurement practices.

5.9.5 The only exceptions to the standard lock and lock identification rule are designed energy control devices like the Kirk Key or Castell Locking Systems.

5.9.6 Written permission from the Health, Safety and Security Department must be obtained before using such an energy control device.

5.9.7 **Personal Safety Lock:** A red, keyed lock having only one key, used only by the trained and authorized employee the lock was assigned to ensure personal protection during lockout/tryout energy isolation activities.

- Employees shall carry their personal safety lock while at work as part of their standard PPE (Personal Protective Equipment).
- Each employee is responsible for keeping the key to their assigned personal safety lock in their possession.

5.9.8 A personal safety lock shall not be applied or removed by anyone else.

Energy Isolation Safety Standard

- 5.9.9 Personal safety locks must be marked with at least the employee's name and one other form of identification such as department name, department identification number, clock/badge number, or specific crew.
- 5.9.10 Employees shall have possession of their personal safety lock key during the entire time they are locked out on a piece of equipment.
- 5.9.11 At no time shall any employee enter a protected area without possession of a key that controls the energy sources.
- 5.9.12 There is no limit to the number of personal safety locks an employee can be issued.
- 5.9.13 Equipment Lock: A blue, keyed lock having only one key, used to secure energy isolation devices on machines or equipment to isolate energy sources following established, written energy isolation verification procedures.
- 5.9.14 Equipment locks shall be engraved or have a durable tag with a grommet attached to the equipment lock shank with terminology matching the energy isolation device (example #5 Scrap Baler Hydraulic Supply Valve).
- 5.9.15 Because energy isolation device terminology can be lengthy, it is permitted to number equipment locks sequentially if a legend exists describing each equipment lock. A legend shall be provided with each energy isolation. The energy isolation and verification procedure should be the legend.
- 5.9.16 The corresponding equipment lock key number must match the equipment lock number. This is important so all elements of the energy isolation can be verified.
- 5.9.17 The equipment lock key number must be visible from the outside of the group lockout box.
- 5.9.18 It is permissible to attach a durable identification tag to the key to be visible outside the group lockout box.
- 5.9.19 An equipment lock must be attached to each personal protective ground wire installed. The lock reminds you that ground wires are applied to crane hot rails or other devices to reduce the likelihood of powering on with the grounds still applied.
- 5.9.20 Control or Cross Lock: A yellow, keyed lock having only one key used to secure the group lockout box to provide continuity of a lockout for multiple energy sources, multiple crews, or across shifts.

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- The control lock is applied to the group lockout box after all equipment lock keys have been placed in the group lockout box. Before applying personal safety locks, the control lock is secured on the group lockout box. The control lock shall not be removed until all personal safety locks have been removed from the box; therefore, the control lock is the first lock on the group lockout box and the last lock to be removed.

5.9.21 All *group lockout boxes* at Cleveland-Cliffs Steel shall have a control lock applied.

5.9.22 *Control locks* shall be engraved with the division name and "Control Lock."

5.9.23 The *control lock* shall be identified on the Locked Key Box Control List by writing "Control Lock" followed by the authorized employee's name who placed the lock.

5.9.24 When a shift change occurs, and the authorized person holding the control lock key needs to leave the facility, that control lock key shall be passed to the oncoming authorized person. In addition, the control lock shall be identified on the Group Lockout Box Control List by writing "Control Lock," followed by the new authorized individual's name who now controls the control lock key. The previous control lock entry on the Locked Key Box Control List shall be dated, time-stamped, and lined out by the outgoing authorized person (see Figure 6 below).

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7:57 PM

GROUP LOCKOUT BOX CONTROL LIST

Job(s): Change RM Edger Rolls

John Doe is the Group Lockout Leader (GLL) for this lockout and has placed a Control Lock on the Group Lockout Box.

BADGE #	EACH INDIVIDUAL'S NAME	DEPARTMENT OR CONTRACTOR	LOCKED		UNLOCKED	
			DATE	TIME	DATE	TIME
34210	John Doe	Hot Mill (Control Lock)	21-Dec	11:00 AM		
9285	Mary Slaughter	Mobile Mtc Dept	21-Dec	11:05 AM	21-Dec	1:46 PM
11411	Frank Bellis	Hot Mill	21-Dec	11:06 AM	21-Dec	1:35 PM
11521	Larry Dauben	Mobile Mtc Dept	21-Dec	11:11 AM	21-Dec	1:30 PM
34210	John Doe	Hot Mill	21-Dec	11:20 AM	21-Dec	1:47 PM
John Doe placed the Control Lock and his Personal Safety Lock. Why Personal Safety Lock? If John Doe leaves the plant, for example, John is going to remove his Personal Safety Lock and pass the key to the Control Lock to the new Group Lockout Leader (GLL) maintaining the lockout/tryout integrity. The Control Lock is the <u>first</u> lock placed and <u>last</u> lock removed from the Group Lockout Box.						

| 21991 | Paul Smith | Hot Mill (Control Lock) | 21-Dec | 11:00 AM | 21-Dec | 2:00 PM |
| **When the GLL holding the Control Lock key needs to leave the facility or is reassigned, that Control Lock key shall be passed to the new GLL. In addition, the previous Control Lock entry shall be lined out and a new Control Lock entry documented.** | | | | | | |

Figure 6 (above) - Group Lockout Box Control List

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- 5.9.25 The passing of the control lock key and documenting the change of control for the control lock key continues as necessary (multiple shift changes) until the lockout/tryout activity is complete.
- 5.9.26 The control lock key is the only key that may be passed from one authorized person to the following authorized person. A control lock shall be used on every lockout activity where a lockbox is utilized.
- 5.9.27 Idled Equipment or Do Not Operate Locks (Green) locks shall be used only to control idled equipment and equipment you do not want someone to operate.
- 5.9.28 Green locks are not to be used to protect personnel for servicing or maintenance.
- 5.9.29 If the machine or equipment is going to be idled beyond a month, apply a green lock to the applicable energy isolation devices.
- If you do not want personnel to use the machine or equipment, apply a green lock and "Do Not Operate" tag.
- 5.9.30 All locks shall be purchased only through the approved safety vendor.

5.10 Lockout Hasp and Group Lockout Box

- 5.10.1 Lockout Hasp: A lockout hasp is a multiple lock device.
- A lockout hasp is ideal when more than one person works on machinery or equipment to isolate a single energy isolation device.
 - The lockout hasp is attached to the equipment's energy isolation device, and each person locking out applies their personal safety lock to the lockout hasp.
 - A control lock is not required for a hasp unless a hasp is used on a group lockout box.
 - If you approach a lockout hasp and only one hole is remaining to place your personal safety lock, avoid applying your personal safety lock; instead, add another lockout hasp to that hole, and then apply your personal safety lock to the newly attached lockout hasp so additional personnel can apply their personal safety lock.
- 5.10.2 There are situations where using a group lockout box is more practical than a lockout hasp, such as when multiple energy sources need to be isolated.

Energy Isolation Safety Standard

5.10.3 Group Lockout Box (*Figure 7, right*): A container with a see-through front used in group lockout to secure all keys to the equipment locks secured on an energy isolating device(s) and permits all exposed persons to apply their personal safety lock.

5.10.4 Group lockout boxes are used when more than one employee and more than one energy source are isolated.

5.10.5 Lockboxes shall be arranged, organized, and labeled regardless of whether they are mounted on the wall or placed near the locked and tried-out machine or equipment to create one convenient location for all involved to identify the lockboxes readily. This arrangement (see Figures 7 above and 8 below) is called a lockout station.



Figure 8 (above) – This is the lockout box you see on the far left in Figure 7. This lockout box has hooks numbered to match each energy-isolation device to lockout equipment MD1. The hooks house an equipment lock (if not locked out) or the key to that equipment lock (if locked out).

5.10.6 A Group lockout box's purpose is to ensure that the equipment lock keys cannot be accessed until the control lock has been removed. The control lock's purpose is to maintain the established lockout/tryout integrity during multiple shifts, shift changes, and throughout the lockout activity.

- **IMPORTANT:** To maintain the integrity of an established lockout/tryout activity, a control lock shall be the first lock placed on a group lockout box after equipment lock keys are placed in the group lockout box and the last lock removed from the group lockout box.

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5.10.7 Group lockout boxes must have a see-through face so that the number of keys and the identity of the keys is visible from outside the box. The see-through group lockout box requirement applies to all group lockout boxes (fixed location and portable) used at Cleveland-Cliffs Steel.

5.10.8 Group lockout boxes must have the Group Lockout Box Control List attached. In addition, control locks shall be engraved with the division name and "Control Lock."

5.11 The following steps must be followed when implementing a *group lockout box*:

- As a job requiring a group lockout box is implemented, you must identify which energy control and verification procedures are necessary. The energy control and verification procedure will then be used to ensure the equipment is locked out and tried out correctly.

5.11.1 Whenever a group lockout box lockout is implemented, a control lock shall be placed on the group lockout box to control access to the equipment lock keys. The control lock shall be the first lock-on and the last lock-off.

5.11.2 The group lockout leader will determine which authorized employee possesses the control lock key (for the shift) for a group lockout box. Control lock keys shall not be left unsecured (e.g., in office drawers, etc.).

5.11.3 The employee in possession of the control lock key is responsible for removing the control lock when:

- The job is complete, and the post energy isolation activity checklist (CCS-HSS-FRM-0005.0) is filled out, or when,
- The lockout is being altered, requiring modification to the energy isolation verification procedure form or a new energy isolation.
- Whenever a lock box is opened and a key(s) removed, it needs to be noted on the energy isolation verification procedure form immediately.

5.11.4 If there are any alterations to an energy isolation activity related to the servicing/maintenance that affects the energy isolation and verification procedure (e.g., testing, changes in what is isolated, and change in scope of servicing/maintenance):

- All changes made to the previously locked and tried-out equipment must be documented on the energy isolation verification procedure (CCS-HSS-FRM-0001.0).
- All personnel affected by the energy isolation activity are notified through a Job Safety Briefing, TAKE 5, or JJHAC.
- All authorized employees re-applying their personal safety lock must re-verify that equipment that may affect their safety is locked out and tried out before returning to work by at least reviewing the energy isolation and verification procedure.
- Do NOT assume you are still protected because you were protected when you first placed your personal safety lock on the group lockout box.

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- 5.11.5 Whenever a group lockout box lockout is implemented, and employees apply their personal safety locks for protection, the Group Lockout Box Control List shall be utilized to document who and when employees are locked out on the group lockout box and when employees remove their personal safety lock from the group lockout box.
- 5.11.6 Two people, at minimum, are required when using the group lockout box system.
- First, one authorized employee [A] shall place the equipment locks.
 - Next, another employee [B] verifies the placement of the equipment lock(s) according to the written energy isolation verification procedure.
 - Then, either employee [A] or [B] shall ensure the equipment is tried out.
 - Tryout can be accomplished through one or more of the following acts:
 - Attempting to operate the equipment
 - Visually verifying that needed safeguards are in place or that blades in the corresponding disconnect box are disengaged
 - Using a voltmeter or other test device
 - Some lockout/tryout activities will require all three tryout methods to ensure that *energy sources* are properly locked out.
- 5.11.7 The identity of the equipment lock keys inside the group lockout box must be visible.
- 5.11.8 A copy of the energy isolation verification procedure shall be placed outside the group lockout box in a document holder attached to the group lockout box for employee viewing.
- 5.11.9 During shift change, the oncoming authorized employees shall ensure that the group lockout box is still in effect before placing their personal safety lock(s) on the group lockout box and starting to work.
- 5.11.10 Before removing the control lock of any lockout/tryout scenario and releasing the equipment to operations, verify all work is complete, and a post energy isolation

Energy Isolation Safety Standard

activity checklist is completed and signed by a Cleveland-Cliffs Steel responsible Manager.

- 5.11.11 The completed post activity checklist (CCS-HSS-FRM-0001.0) must be returned to the Health, Safety, and Security Department for auditing.
- 5.11.12 Any time a personal safety lock or equipment lock is placed on an energy isolation device (e.g., disconnect switch, valve, pin, etc.), it shall be locked in a manner to prevent removal without the use of excessive force or unusual techniques.
- 5.11.13 Locks shall never be left on the energy isolation device, group lockout box, or lockout hasp while un-locked; they shall be placed and immediately locked.
- 5.11.14 Equipment locks shall be returned to the designated storage location while not being used for lockout.

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- 5.11.15 Keys for all equipment locks placed for lockout/tryout shall always be controlled. Equipment lock keys shall never be left unsecured in the group lockout box.
- 5.11.16 Each energy isolation device on the equipment is secured with an assigned equipment lock by the Group Lockout Leader per the written energy isolation and verification procedure.
- 5.11.17 An authorized employee(s) places each equipment lock key in the group lockout box after securing each equipment lock on the assigned energy isolation device.
- 5.11.18 An assigned control lock is applied to the group lockout box by the Group Lockout Leader.
- 5.11.19 The assigned control lock is the first lock placed on a group lockout box and the last lock removed from the group lockout box.
- 5.11.20 Each authorized employee places their personal safety lock on the group lockout box established for the equipment to be serviced.
 - **IMPORTANT:** You must verify that you are placing your personal safety lock on the correct group lockout box or boxes.
- 5.11.21 When work requiring lockout has been completed, the authorized employees remove their personal safety lock from the group lockout box.
- 5.11.22 When energy isolation work is complete, the Group Lockout Leader completes a post activity checklist (CCS-HSS-FRM-0005.0).
- 5.11.23 Next, the Group Lockout Leader removes the control lock from the group lockout box, freeing the keys to the equipment locks.
- 5.11.24 For the Kirk Key and Castell Control Systems, follow the Risk Assessment, Practicality, and Justification requirements.

5.12 Energy Control Verification Procedures

- 5.12.1 Written energy control verification procedures shall be developed and maintained in each department.
- 5.12.2 Energy control verification procedures shall cover the steps to ensure that the serviced equipment is isolated and secured from all energy sources, including stored or residual energy, which could injure personnel working on the equipment.
- 5.12.3 Energy control verification procedures shall exist to lockout and try out machines, equipment, or operating units and be task specific.

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- 5.12.4 When using an energy control and verification procedure not written explicitly for a servicing or maintenance job/task, all energy-isolation devices not needed to protect authorized or affected employees shall be lined out and initialed on the energy isolation and verification procedure.

5.13 Written energy control and verification procedures shall identify and provide information on the following:

- 5.13.1 A specific statement of the intended use of the procedures.
- 5.13.2 Type of machinery or equipment being locked out.
- 5.13.3 Types of energy sources and magnitude, such as electrical (480v), pneumatic (35 psi), hydraulic, steam, etc., must be locked out.
- 5.13.4 Method and location of the lockout.
- 5.13.5 Location(s) where tryout takes place.
- 5.13.6 The stored or residual energy sources such as capacitors, springs, elevated machine members, rotating flywheels, pressure in hydraulic, pneumatic, or steam systems, etc., must be controlled.
- 5.13.7 Method and location to dissipate or restrain by grounding, repositioning, blocking, pinning, bleeding down, etc.
- 5.13.8 If applicable, the method and location of lockout of the dissipating or restraining device.
- 5.13.9 Method and location where tryout will take place.
- 5.13.10 The method of verification that shall be used to assure that a Zero Energy State has been achieved and maintained for the machinery or equipment being serviced.
- 5.13.11 Lockout involving just one authorized individual for a single energy source requires a "Tryout" of the equipment but does not require a written energy control and verification procedure.

5.14 Troubleshooting, Testing, or Positioning Equipment/Machines

- 5.14.1 In situations where lockout devices must be removed temporarily from the energy isolating devices and the machine or equipment energized to troubleshoot, test, or position the equipment or machine, the following must be observed:

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- Use three-part communication.
- Survey and replace any missing machine guarding and clear the machine of tools and equipment that may interfere with its operation.
- Make certain controls are in the neutral or "OFF" position.
- All employees (affected, authorized, and others) are notified that the lockout devices will be removed and the equipment reenergized. All employees shall be located outside of the radius of movement of the equipment and in clear view of the point of control of the machine area.
- Remove the lockout device(s) for the equipment sections to be checked.
- Energize and proceed with the equipment's troubleshooting, testing, or positioning only while directly communicating with all the above employees.
- When finished, shut down the equipment or machine, de-energize all energy sources, replace the lockout devices, and test operating controls.
- Document the changes to the energy control and verification procedure immediately.
 - CAUTION: Automatic equipment has a built-in memory and, in many instances, may cycle or be activated when power is restored to equipment.

5.14.2 Where equipment must be operated at intervals to complete repair work, energy control and verification procedures must be established to remove and replace lockout devices.

5.14.3 There must be three-part communications between the persons removing the lock and operating the equipment and those at the equipment who may be aligning or performing other work in conjunction with the repair.

5.15 Energy control of Piping Systems

5.15.1 Before any piping system is opened, energy control and verification procedures must be established to safely reduce to a zero-energy state and verify that this has been done for the section being worked on.

5.15.2 System shut down to zero energy state is always preferred. However, if the work to be done will affect the integrity of the system valves, blanks, blinds, or other controls, the system must be at zero energy.

5.15.3 Blank, Blind, or bolted slip blind is acceptable if there has been no failure under the same or like conditions.

5.15.4 Conditions for the use of Double Block and Bleed or Double Block and Vent:

- The bleed or vent has been confirmed to be open visually, mechanically (rod), or by the flow.
- Release from the bleed or vent will create no hazard.
- System pressure will not exceed the rating. Piping system rated by pressure and temperature (most significant for steam).
- The bleed-off vent is large enough to release pressure.

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5.15.5 Conditions under which use of double block and bleed is not permitted:

- There has been a previous failure under the same or like conditions.
- System pressure is not known.
- Piping rating not known.
- The piping system is not rated.
- The system is susceptible to hammering or other uncontrolled pressure surges.
- There is no procedure for bringing a system up to pressure during startup.
- There have been changes that could affect the flow of material made to the system since its last operation.

5.15.6 Conditions for the use of a Single Valve:

- System pressure will not exceed 75% of the rating. Piping system rated by pressure and temperature (most significant for steam).
- No defects or problems are detected by careful inspection of the valve.
- Stored energy and energy potential are extremely low/minimal, and the maximum system pressure does not exceed the rating. Piping system rated by pressure and temperature (Most significant for steam).
- The material being valved off creates no significant hazard.

5.15.7 Conditions under which use of a Single Valve is not permitted:

- There has been a previous failure under the same or like conditions.
- System pressure is not known.
- Valve rating is not known, or the valve is not rated.
- The system is susceptible to hammering or other uncontrolled pressure surges.
- There is no procedure for bringing a system up to pressure during startup.
- Significant changes have been made to the system since its last operation.
- When the system contains acids, caustic or poisonous material of such strength or concentration, that contact can cause injury.
- The Confined Space Standard requires blank, blind, spool, mis-align, plug or double block, and bleed for isolation.

5.15.8 Start-Up of Equipment – A thorough inspection must be conducted to ensure all persons exposed to stored energy safety hazards are informed that the equipment is to be started.

5.15.9 Ensure all employees are removed from the exposure before starting the equipment.

5.15.10 Specify any other necessary steps which must be taken before equipment is started.

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5.16 Periodic Inspections of energy control and Verification Procedures and Alternative Controls

- 5.16.1 Periodic inspections of energy isolation and verification procedures and alternative controls shall be conducted at least once per year, and consist of the following required elements:
- 5.16.2 Shall be performed by an authorized employee other than the ones utilizing the energy control and verification procedure or alternative measure(s) being inspected.
- 5.16.3 Shall be conducted to correct any deviations or inadequacies identified during the inspection.
- 5.16.4 Shall include a review between the inspector and each authorized employee of that employee's responsibilities under the inspected energy isolation and verification procedure.
- 5.16.5 The periodic inspection findings shall be documented using the periodic inspections: energy isolation and verification procedure and alternative Control Form, CCS-HSS-FRM-0006.0.
- 5.16.6 All Operating Section and Shift Managers will be required to complete one periodic inspection of an alternative control per week where applicable.
- 5.16.7 All Maintenance Section and Shift Managers must complete one periodic inspection of an energy isolation and verification procedure activity for every standard downturn and outage.
- 5.16.8 All plant Safety Personnel will be required to complete one periodic inspection of alternative control or energy isolation and verification procedure activity per week.
- 5.16.9 You may group distinct energy isolation and verification procedures associated with similar machines or equipment and consider the group of distinct energy isolation and verification procedures to be a single procedure for conducting a periodic inspection if the machines or equipment in the group have the same or similar types of control measures.
- 5.16.10 You may elect to group energy isolation and verification procedures described above and then inspect a representative number of such employees implementing one procedure within each group.
- 5.16.11 This approach is acceptable if the inspection sampling reflects division servicing and maintenance operations and hazardous energy control practices for the inspected energy isolation and verification procedure.

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5.16.12 The periodic inspections must contain at least two components:

- An inspection of each energy isolation and verification procedure (or grouping), and
- A review of each employee's responsibilities under the energy isolation and verification procedure being inspected.

5.16.13 Each energy control and verification procedure (or grouping) must be inspected annually to ensure that the lockout/tryout verification program is correctly utilized.

5.16.14 These inspections must demonstrate the energy isolation and verification procedures and must be performed while authorized employees perform servicing and maintenance activities on machines or equipment.

5.16.15 The employee performing the periodic inspection does not have to observe every authorized employee implementing the energy isolation and verification procedure on the machine or equipment they are authorized to perform servicing and maintenance to meet the requirements. The inspector participating in the review when the lockout is used needs to:

- Observe a representative number of such employees while implementing the energy isolation and verification procedure, and
- Talk with all other authorized employees even though they may not implement the energy isolation and verification procedure.

5.16.16 The periodic inspection is not required to be conducted in separate one-on-one meetings; it can involve the inspector meeting simultaneously with a group of authorized employees. Group meetings can be one of the most effective methods to

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conduct the inspection and clarify to the employees that they must follow the energy isolation and verification procedure carefully.

5.17 Operating, Repairing, Servicing, and Inspection of Equipment

- 5.17.1 Only a trained operator or maintenance person may energize/de-energize and operate a machine.
- 5.17.2 Do not operate, repair or test any machinery, electrical apparatus, or other equipment unless part of your assigned duties.
- 5.17.3 Do not lubricate, clean, repair, inspect or operate a machine you do not understand. When in doubt, ask your manager.
- 5.17.4 If you are authorized to operate machinery, follow all safety procedures, including the JSA, JSHA (Job Safety and Health Analysis), Department Rules, and Manufacturer's recommendations established for the machine or equipment.
- 5.17.5 Authorized Cleveland-Cliffs Steel employees must perform the energy isolation and verification procedures including de-energizing, lockout, verification, and tryout before servicing and maintaining machines and equipment.
- 5.17.6 All employees shall be trained on the energy isolation and verification procedures to ensure that each employee understands the purpose and function of this energy

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isolation standard and the knowledge and skills required for safe application, usage, and removal of energy controls.

- 5.17.7 Operators of machines, cranes, and other equipment must secure the equipment under the established energy isolation and verification procedure or other procedure before leaving the job unless relieved on the job by another operator.
- 5.17.8 Any unsafe machine to operate must be de-energized and locked out. Notify a manager of any broken or unsafe machinery or equipment.
- 5.17.9 When opening an electric safety or disconnect switch:
- Do not open under load
 - Stand to the side of the disconnect box, not in front of it
 - Turn face away from switch (reduce risk of being blinded or damage to breathing)
 - Hold breath (so if arc happens, you do not inhale liquid metal)
 - Close eyes (so if arc happens, you are not blinded)
 - Throw switch
- 5.17.10 Contact your manager if it is impossible to stand clear from the front of the box.
- 5.17.11 Only electrically qualified and authorized personnel are permitted to open a disconnect switch to verify that the knife blades are in the open position.
- 5.17.12 Do not remove any personal safety lock unless you place it. Remove only your lock only after you are sure all personnel is in the clear.
- 5.17.13 Do not remove a DANGER or CAUTION sign unless you placed it, and then only after you are sure the dangerous condition has been corrected.
- 5.17.14 Contact your manager if a lock must be removed and the person who placed it cannot be found. The lock shall be removed by following Safety Lock Removal Incident Report Form (CCS-HSS-FRM-0007.0).

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- 5.17.15 Only authorized personnel can adjust machinery in motion with a written procedure approved by the plant General Manager and Health, Safety, and Security Manager.
 - If you are authorized to make minor production adjustments on machines, follow the procedures that have been approved.
- 5.17.16 Be sure to guard against the movement of parts or machinery which might injure you or others. Energy control and verification procedures must be followed during the service or maintenance of the equipment.
- 5.17.17 Machinery must not be operated if guards have been removed for repairs or adjustments.
- 5.17.18 Do not remove locks or energize equipment until all guards and safety devices are correctly replaced, secured, and operable.
- 5.17.19 Machine guarding and safety devices shall be affixed to the machine or secured elsewhere as designed by the equipment manufacturer and approved by completing an Equipment Safety Inspection and Startup Procedure Form.
- 5.17.20 Pressure must be bled off before repair or service work on hydraulic, steam, gas, air, oil, water lines, or connected equipment.
 - Flow valves must be closed and locked, and pressure relief valves must be locked in the open position.
 - Accumulators must be discharged.
- 5.17.21 Valve lockouts must be made following energy control and verification procedures before any repair or service.
- 5.17.22 Before the flow of steam, gas, air, oil, or water is restored to a system, the person doing so must ensure that no one can be injured as the systems are activated.
- 5.17.23 Use extreme caution when working around conveyor systems, especially conveyor belts.
- 5.17.24 Do not shovel onto moving belts unless authorized by specific written procedures and all guarding is in place.
- 5.17.25 Make sure the emergency stop is functioning correctly.
- 5.17.26 Moving rolls shall not be touched by hand at any time.
- 5.17.27 Work rolls shall not be cleaned or polished by hand in the mill and under power.
- 5.17.28 If work rolls are installed in the mill and must be cleaned or polished by hand, they shall be locked out.

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5.17.29 Work rolls installed in the mill and moving or under power may be cleaned or polished under the following conditions:

- Tools shall be provided to facilitate the cleaning or polishing of work rolls with the hands at a safe distance from the moving rolls.
- A procedure directs using a tool(s) to keep hands away from the moving rolls.
- The cleaning or polishing of the work rolls shall be performed at the side of the mill opposite the work roll's in-running nip point. If necessary, the direction of rotation of the work rolls may be reversed to eliminate the in-running nip-point hazard.

5.17.30 The strip shall not be touched with any body part when moving under tension.

- The strip may be positioned or worked on using extension/hands-free tools when the line is stopped or in jog mode.
- Extension/hands-free tools shall be used when clearing jams and removing scrap from the line unless the scrap has been cut into easily manageable, i.e., sample width or smaller, pieces.
- A variance to this rule may be granted by the Plant General Manager and the Health, Safety, and Security Manager after all alternatives have been exhausted. In addition, a written safe procedure for performing the job must be developed.

5.18 LOTO for Industrial Vehicles

5.18.1 Any vehicle has the potential to contain various types of hazardous energy:

- Mechanical hazards associated with start-up.
- Mechanical motions due to moving power transmission components.
- Mechanical hazards with disc brake springs and tire components.
- Hydraulic hazards associated with fluid pressure and loss.
- Gravitational energy from elevated vehicles and lifts.
- Gravitational energy from elevated truck-mounted equipment such as dump beds.
- Airbag explosion.
- Electric battery shock, arc, or burns.
- Fuel and fluid system fire or explosion.
- Hot or cryogenic fluid and surface hazards.
- Chemical energy (e.g., battery acid, coolants, lubricants, etc.).

5.18.2 Before servicing an industrial vehicle, an initial step is to remove the ignition key from the ignition switch.

- The repairperson shall maintain sole possession of that key, placing it in their pocket or a group lockout box if more than one person is servicing the vehicle.
- Doors to the vehicle should also be locked.

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5.18.3 If the above two steps are insufficient to prevent an engine from accidentally starting, the repair person shall disconnect the battery cables.

- Disconnecting batteries is essential when working on or around cooling fans, which can automatically start up after removing the ignition key.
- It is good to allow an engine to cool down sufficiently. Doing so will allow heat and subsequent pressure to dissipate, helping control those forms of hazardous energy.

5.18.4 Depending on the type of vehicle being worked on and the service being performed, the repair person must either lower or block-up components held up by hydraulic pressure, such as dump beds and aerial lifts. It may also be necessary to chock or block wheels to prevent a vehicle from rolling.

5.18.5 Dump beds can be incredibly perplexing. Manufacturer-provided props often do not provide the clearance necessary to perform many under-bed repairs. This incites some repair persons to create their support braces, many of which do not provide adequate protection.

- The equipment manufacturer shall be contacted to discuss an alternative body prop or another suitable method to perform the work.
- Remember that reenergizing a vehicle or component is often necessary for testing or troubleshooting.
- A procedure that defines the sequence of actions necessary to accomplish the task safely shall be developed.

5.18.6 It is imperative to check its service guidelines with the specific industrial vehicle manufacturer. Those service guidelines can double as guidelines to help develop energy isolation and verification procedures.

5.18.7 Thorough energy isolation and verification procedures must be developed for shop equipment, including lifts, jacks, tire changers, wheel balancers, and pneumatic (compressed air) systems.

- Procedures should be aligned with the equipment manufacturers' service guidelines.
- It is essential that the industrial vehicle energy isolation and training program helps to ensure that employees understand all relevant energy sources, methods of isolating or controlling those energy sources, and methods to verify effective energy control. Therefore, employee retraining may be necessary from time to time.

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5.19 Tagout Alternative/ Tagout System Limitations

- 5.19.1 A tag must only be used instead of an equipment lock when there is no means to apply a lock.
- 5.19.2 If a machine cannot be locked out, it must be tagged. If energy isolating devices cannot accommodate locks, a tag shall be used to warn employees that a machine is de-energized for servicing.
- 5.19.3 The procedure for installing and removing tags is the same as the lockout procedure.
- 5.19.4 Tags are only a warning device affixed to energy isolating devices and do not provide the physical restraint on those devices provided by a lock.
- 5.19.5 When a tag is attached to an energy isolating means, it is not to be removed without authorization from the authorized person responsible. It is also never to be bypassed, ignored, or otherwise defeated.
- 5.19.6 Tags must be legible and understandable by all authorized employees, affected employees, and all other employees whose work operations are or may be in the area to be effective.
- 5.19.7 Tags and their means of attachment must be made of materials that will withstand the environmental conditions encountered in the workplace.
- 5.19.8 Employees shall be made to understand that tags may evoke a false sense of security and their meaning need to be understood as part of the overall energy control program.
- 5.19.9 Tags must be securely fastened to the energy isolating devices so they cannot be inadvertently or accidentally detached during use.
- 5.19.10 Tags shall include the employee's name, division, and date.
- 5.19.11 Like a lock, the tag must be securely fastened, in a position to be seen by all employees, and standardized. Therefore, the following safety tags (Figure 9) have been standardized, and no others are permitted to be used:



Figure 9 (Above) - Approved Durable Tag with Grommet

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5.20 Alternative Control - Requirements

5.20.1 You cannot use control circuit type devices such as start/stop push buttons, e-stops, selector switches, presence sensing devices, or limiting switches instead of energy-isolating devices for equipment lockout. This is because these control circuit type devices do not physically isolate equipment from the equipment's energy source. As a result, unanticipated circumstances such as welded contacts or a loose wire shorting out can cause unexpected energization or startup.

5.20.2 Under the "minor servicing," circuitry meeting the control reliability requirements of ANSI Z244.1-2016 provides the alternative control measures required to control hazardous energy.

- Control reliability is the capability of the control system and related interfacing to achieve a safe state in the event of a failure.
 - For example, an interlock or e-stop having a dual-channel (redundant) circuit with a self-monitoring safety module typically meets the control reliability requirements.
 - A basic start/stop circuit is equipped with a single-contact button (single channel), even though it may be "safety-rated" (positive contacts).
 - Due to the number of times the e-stop is cycled and the simplicity of the circuit, there are still several chances for a single component to fail, leading to circuit failure.
 - Because of the lack of redundancy and self-checking circuitry, this single-channel e-stop circuit is not considered control reliable.
- If you are considering incorporating control reliable safety control circuits, a thorough practicability/justification analysis (CCS-HSS-FRM-0009.0), risk assessment (CCS-HSS-FRM-0010.0), and other alternative control evaluations (see Appendix A) must be done to document and verify the necessary integrity and performance levels of the control system. This ensures you are providing equivalent and effective employee protection.
- The machine supplier, competent machine integrator, or internal engineering delegate can only make the above control reliable safety control circuits determination. These persons can determine a machine's performance level according to (EN) ISO standards which correspond to the machine performance level expectations. Machines with interlocks, e-stops, or presence sensing technologies have an up-to-date (no significant changes made to machine or wiring) machine performance level review on file.

5.20.3 The exemption for minor tool changes and adjustments and other minor servicing activities is a unique situation and shall not be used to avoid lockout/tryout.

5.20.4 Lockout/tryout shall be used unless the division manager can demonstrate to the Health, Safety, and Security Manager that an alternative control provides equal or greater protection. When energy isolation is not used, approved alternative control shall be used.

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5.20.5 There are three specific criteria to determine if the minor servicing exception would apply to a particular activity:

- The activity must be conducted during normal production operations.
- The activity must be:
 - Routine: The activity must be a regular course of procedure and follow established practices.
 - Repetitive: The activity must be regularly repeated as part of the production process.
 - Integral: The activity must be essential to the production process.

5.20.6 Before alternative controls are used, the following shall be completed and approved per ANSI Z244.1-2016, The Control of Hazardous Energy Lockout and Alternative Controls:

- A practicability/justification analysis per clause 8.2.1 (CCS-HSS-FRM-0009.0).
- A risk assessment as per clause 8.2.2 (CCS-HSS-FRM-0010.0).
- Other alternative control evaluations as described in clauses 8.2.3 – 8.2.14 (See Appendix A).

5.20.7 Alternative controls shall be added to the safe job procedure (JSHA or JSA) if approved.

5.20.8 There shall be a master list in each division of these approved alternative controls.

5.20.9 Alternative controls must be under the exclusive control of the employee performing the minor servicing.

5.20.10 These alternative controls must enable the employee to safely perform the servicing task without being exposed to the energization or activation of the equipment or the release of stored energy.

5.20.11 If the minor servicing exception criteria are not met in full, this Standard is applicable, and the equipment or machinery must be de-energized. In addition, all potentially hazardous energy must be brought to a zero-energy state.

5.21 Contract Workers (Contractors, Sub-Contractors, Service Providers)

5.21.1 When contractors work at a Cleveland-Cliffs Steel facility, they must follow their written lockout/tryout standard and adhere to the Cleveland-Cliffs Steel Energy Isolation Standard.

5.21.2 All contract workers must comply with these requirements, beginning with, but not limited to, having a red personal safety lock and lock identification.

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5.21.3 Contract workers shall identify personal safety locks with employee and contractor company names.

- Contract workers are permitted to use a durable tag with a grommet attached to the shank of their personal safety lock due to outages and utilization of travelers.

5.21.4 Contract workers are not permitted to operate any energy isolation devices. Cleveland-Cliffs Steel employees shall perform this task. A variance to this rule may be granted by the Plant General Manager and Health, Safety, and Security Manager after written safe procedure for performing the job are submitted and approved.

5.21.5 This Energy Isolation Safety Standard will be distributed to our current contract workforce base through ISNetworld. However, each steel plant's Health, Safety, and Security Department will be required to conduct or administer a face-to-face meeting on the subject with each contract company's representatives to:

- Verify each contract company working at your facility has received a copy of the Standard.
- Answer or record (to later answer) any comments, questions, or concerns.
- Ensure (as much as possible) that each contract company is or will be conducting training on this Standard with their contract employees.

5.21.6 Each facility's employee/management committee will determine how to address training on this Standard for vendors, visitors, and contract workers not associated with ISNetworld.

6 Forms and Documented Information

6.1 Documentation

- 6.1.1 All activity defined within this Lockout/Tryout Standard shall be appropriately documented.
- 6.1.2 All employees' lockout/tryout training activities shall be recorded in your facility's training database or another system.

6.2 Energy Control Verification Procedure (CCS-HSS-FRM-0001.0)

- 6.2.1 A written procedure that describes the verification steps to control the hazardous energy for each piece of equipment, industrial vehicle, or machinery.
- 6.2.2 Maintain current records for the life of the machine or equipment.

6.3 Group Lockout Box Control List (CCS-HSS-FRM-0002.0)

- 6.3.1 A form for documenting authorized employees' and contract workers' names who placed or removed their personal safety lock from a group lockout box.
- 6.3.2 Maintain the current record until all locks on this list are removed.

6.4 Group Lockout Leader Evaluation Checklist (CCS-HSS-FRM-0003.0)

- 6.4.1 A form for evaluating a potential Group Lockout Leader.
- 6.4.2 Maintain record for employment of the employee.

6.5 Training Log (CCS-HSS-FRM-0004.0)

- 6.5.1 This training log is provided to record all training completed that is not documented by other written means such as a completion certificate or other written documentation. It can be used for observational, interaction, classroom, or hands-on training.
- 6.5.2 There shall be a master list maintained in each division of authorized employees, group lockout leaders, second-person verifiers, and testers. The Training Log shall serve as the master list to manage training completion versus trying to find someone's individual record if their training status is questioned.
- 6.5.3 Maintain record for employment of the employees.

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6.6 POST Activity Checklist (CCS-HSS-FRM-0005.0)

- 6.6.1 Before removing the control lock from a group lockout box after maintenance and servicing are complete, this checklist must be completed and signed by a Cleveland-Cliffs Steel responsible manager before releasing the equipment to operations.
- 6.6.2 Maintain record for 1 year + current.

6.7 Periodic Inspections of Energy Isolation Procedure & Alternative Controls (CCS-HSS-FRM-0006.0)

- 6.7.1 Periodic Inspections are a requirement of OSHA and Cleveland-Cliffs Steel. This form is to help you evaluate compliance with the following requirements:
 - 6.7.1.1 Determine whether:
 - The steps in the energy control procedure are being followed.
 - The employees involved know their responsibilities under the procedure.
 - The procedure is adequate to provide the necessary protection and what changes are needed.
 - 6.7.1.2 Emphasize that the person performing the periodic inspection is an authorized employee (other than the one(s) utilizing the procedure being inspected).
 - 6.7.1.3 Determine if any deviations or inadequacies exist.
 - All deviations or inadequacies discovered by the inspections must be corrected and documented.
 - 6.7.1.4 Maintain records for 1 year + current.

6.8 Safety Lock Removal Incident Form (CCS-HSS-FRM-0007.0)

- 6.8.1 A safety lock removal incident form should only be used if the person who initiated the energy isolation cannot complete the lockout process and remove their lock.
- 6.8.2 The decision for safety lock removal is made by management.
- 6.8.3 Safety locks are to be removed **ONLY** by the division manager or a trained person authorized by the division manager in their absence.
- 6.8.4 For record-keeping purposes, complete a Safety Lock Removal Incident Form.
- 6.8.5 Maintain records for 1 year + current.

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6.9 Energy Isolation Personal Protective Grounding Log (CCS-HSS-FRM-0008.0)

- 6.9.1 A form used to track installation and removal of personal protective grounding wires.
- 6.9.2 Personal protective grounds go by several names in the industry: "temporary protective grounds," "ground sets," "ground clusters," or plain "grounds." Personal protective grounds are used whenever workers perform tasks on electrical power systems that may become reenergized for some reason by the reclosing of switches or circuit breakers, static voltages, and induced voltages in outdoor substations or lines and capacitive discharges.
- 6.9.3 The use of personal protective grounding is covered by OSHA 1910.269(n), "Grounding for the Protection of Employees," and the NFPA (National Fire Protection Association) 70E Section 120.3, "Temporary Protective Grounding." Both sources contain very similar requirements.
- 6.9.4 Maintain the current record until all grounds on this log are removed.

6.10 Alternative Control – Practicability/Justification Form (CCS-HSS-FRM-0009.0)

- 6.10.1 The practicability and justification analysis results in a "go" or "no go" decision to lockout.
- 6.10.2 If the practicability or justification analysis results demonstrate that a lockout is a viable option, then lockout shall be used. If the results indicate that lockout is not feasible and an alternative control is suitable for the application, continue to the Risk Assessment (ANSI/ASSP Z244.1) Form in CCS-HSS-FRM-0010.0.
- 6.10.3 Maintain the current record for the life of the machine or equipment.

6.11 Alternative Control – Risk Assessment Form (CCS-HSS-FRM-0010.0)

- 6.11.1 Perform a Risk Assessment. Here you will identify Risk Factors, Hazard(s), Risks, Control Measures, and Alternative Control Justification.
- 6.11.2 Maintain the current record for the life of the machine or equipment.

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6.12 Appendix A – Other Applicable Evaluations for Alternative Control

6.12.1 Industry Best Practices

- Industry best practices represent a prevailing industry way of controlling hazardous energy.

6.12.2 Architecture/Structure

- A key consideration in evaluating the safety performance (reliability) of the alternative control involves the architecture or structure of the system. If components are used that are less than well-tried or less than well-tried designs are implemented, the architecture/structure can significantly improve the system's reliability.

6.12.3 Well-Tried Components

- A well-tried component is a component that is either widely used in the past with successful results in similar applications or made and verified by the component manufacturer using principles that demonstrate its suitability and reliability for safety-related applications.
- The components used can be checked and verified on new machinery and equipment and existing equipment used in the field. Thus, both users and suppliers of machinery can verify the design of existing machinery in the field.

6.12.4 Well-Tried Designs

- A well-tried design is a system that has been demonstrated, based on an analysis of operational experience, for a specific configuration of an element to show that the likelihood of dangerous systematic faults is acceptably low.

6.12.5 Common Cause Failure (CCF)

- Common cause failures refer to a failure resulting from one or more events, causing concurrent failures of two or more separate elements in a redundant architecture/structure subsystem, leading to a dangerous failure.
- A CCF usually results in two or more failures from a single event or cause. Therefore, the design should be assessed to establish the effectiveness of the measures used to safeguard against CCF.

6.12.6 Fault Tolerance

- Fault tolerance pertains to the ability of an alternative control to achieve a safe state in the event of a fault. Faults can include leaking valves, electrical shorts, contactors welding shut, mechanical fractures/breakage, overheating components, and environmental exposure.
- Fault tolerance is determined by the circuit's architecture and is defined as categories and performance levels in ISO 13849-1 or "Control Reliability" as defined in ANSI B11.19.

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6.12.7 Exclusivity or Individual Control

- Examples include, but are not limited to:
 - Locks
 - Size of working zone
 - Proximity to control stations, pendants, or devices
 - Locking devices
 - Passwords
 - Enabling devices
 - Hold to run controls
 - Trapped key systems
 - Presence sensing devices (under some conditions)
 - Interlock blocking device
 - RFID systems
 - Factors that reduce exclusivity shall be considered in alternate risk reduction measures
- Examples include, but are not limited to:
 - Key reproducibility
 - Manual manipulation of controls or devices
 - Password protection and management
 - Visibility of the hazardous area from the reset location (line of sight)

6.12.8 Tamper Resistance

- Whereas exclusivity pertains to a person's ability to control the release of energy, tamper resistance refers to the ability of a person to tamper intentionally, resulting in a system inadvertently energizing or releasing energy. A lock exhibits good tamper resistance due to the physical properties of the lock. Tagout has less tamper resistance as defeating a plastic zip tie requires less force and more easily obtained tools (for example, pliers versus bolt cutters). A warning or tag exhibits even less tamper resistance.
- Selection of system components may also serve to resist tampering (bypass, jumpering, defeat of interlocks, etc.).
- The physical layout of the equipment may achieve tamper resistance. For example, restricting access to areas is another form of physical layout that improves tamper resistance.
- The physical configuration of components may also achieve tamper resistance. Designs that limit or prohibit mechanical manipulation of the controls improve tamper resistance.
- Tamper resistance need not be physical. For example, passwords that restrict access represent another form of tamper resistance.
- Another form of tamper resistance is technical sophistication or complexity. Tamper resistance is improved if bypassing a system requires having access to, and being able to read, the system schematics. Systems requiring specialized skills to operate have greater tamper resistance than those without.

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6.12.9 Program to Support

- The documentation of the tasks which may be performed using a specific alternative control is intended to prevent task creep.

6.12.10 Procedures in Place

- The procedures in place to support an alternative control shall be evaluated. Alternative controls shall include developing, documenting, and using procedures when affected persons are engaged in activities requiring the control of hazardous energy.

6.12.11 Periodic Check and Test

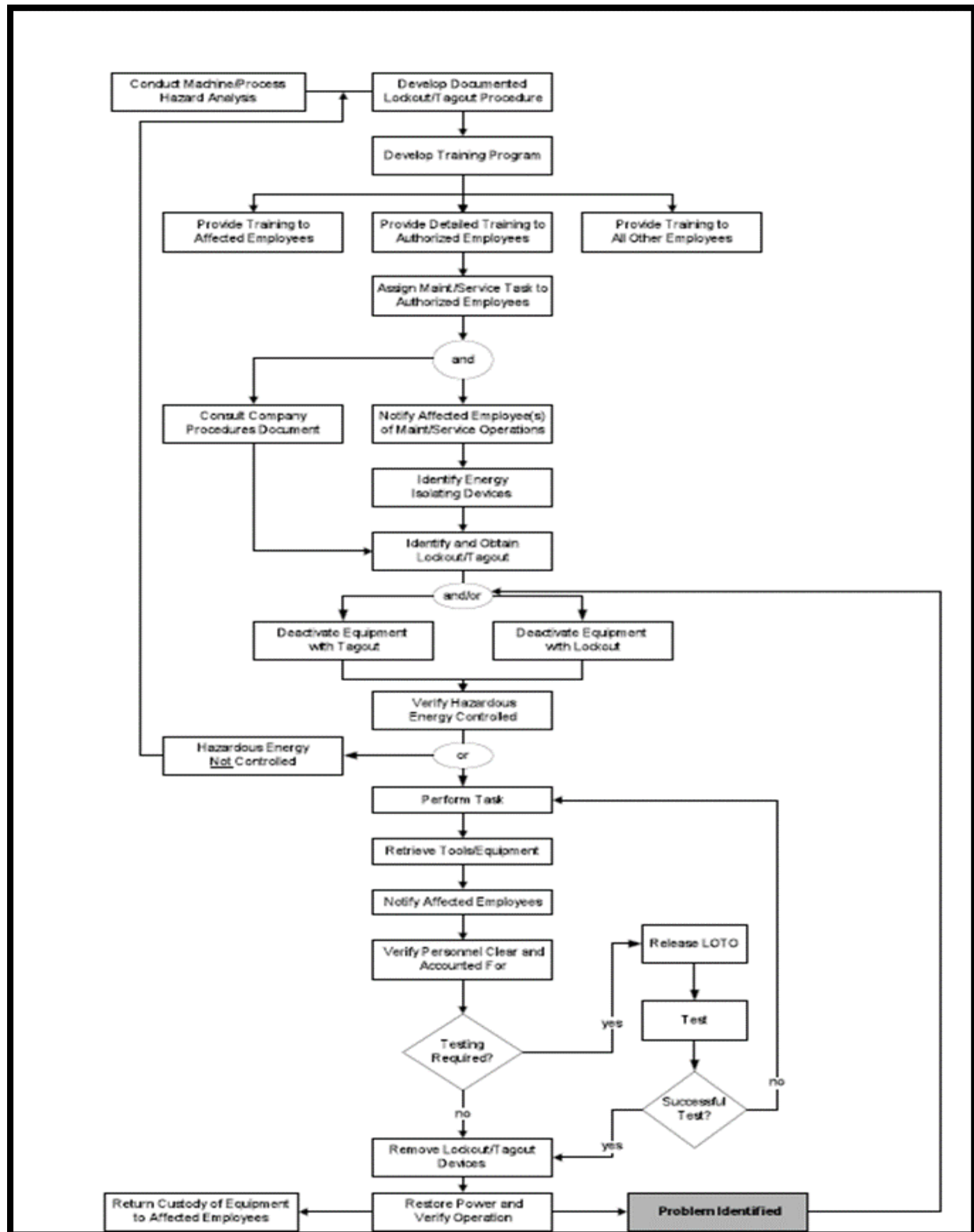
- Where alternative controls are used to control hazardous energy, the need for checking is especially important because, over time, equipment, operations, and work methods can change in small and large ways. Without a program that requires periodic checking and testing, alternative controls can become outdated to the point that they no longer provide adequate protection.
- Checking and testing shall also include correcting any deviations or inadequacies identified.

6.12.12 Review by a Qualified Person

- The persons developing the alternative control should use due diligence throughout the process to achieve high confidence in the results. Confidence can be improved by consulting others with the appropriate knowledge and expertise and having other qualified persons review the design.

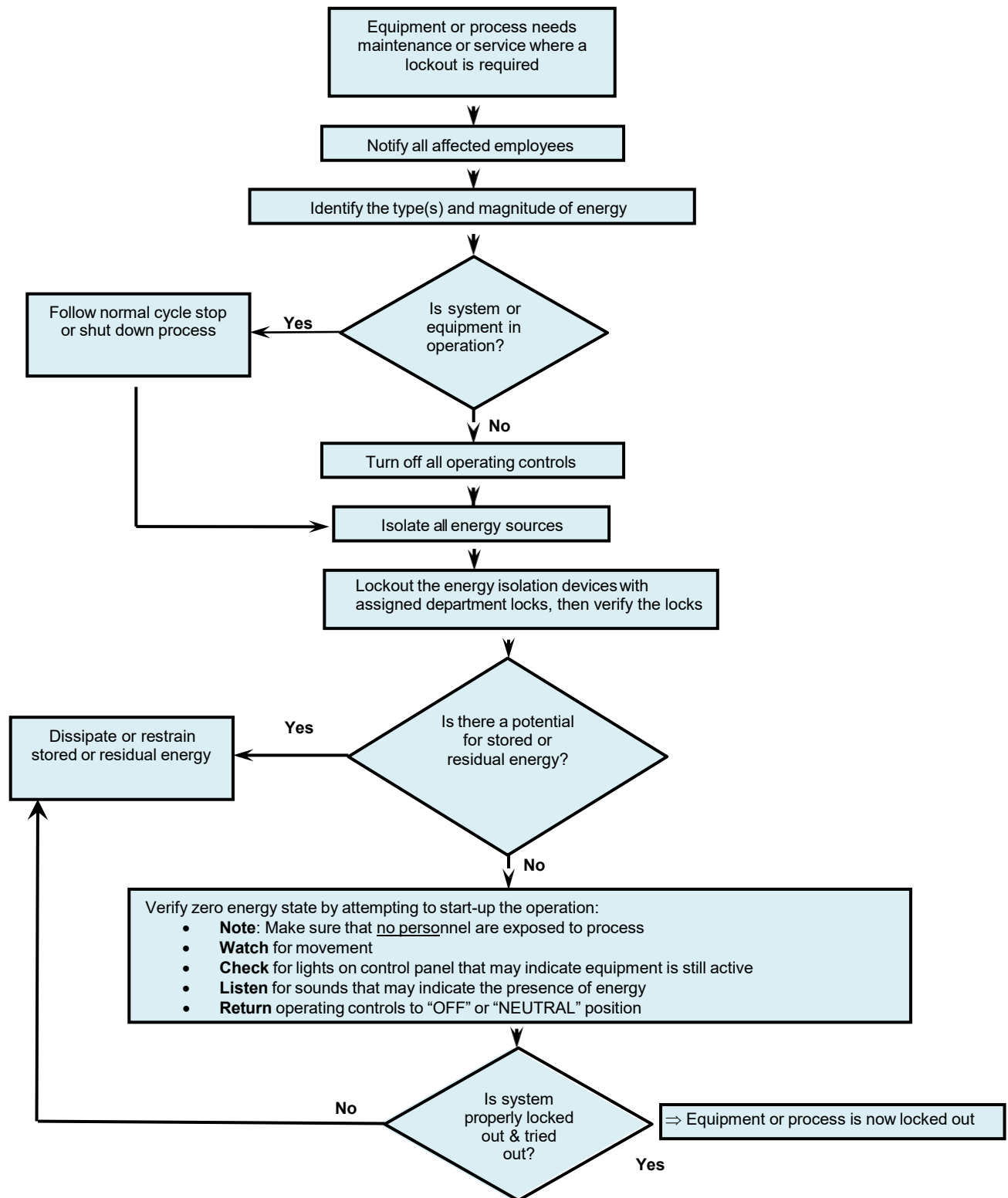
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6.13 Appendix B – Lockout/Tryout Implementation Flow Diagram



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6.14 Appendix C – Functional Lockout Flowchart



7 Opportunities and Risks

8 Revision History

Rev.	Date	Section	Paragraph	Summary of change	Authorized by
0	9-1-2022	Harmonized Standard Requirements and Template	Harmonized Standard Requirements and Template	The initial issue of the CCS Standard for Energy Control	K. D. Berard